Machine Learning-Based Virtual Try-On System for Online Fashion Retail.

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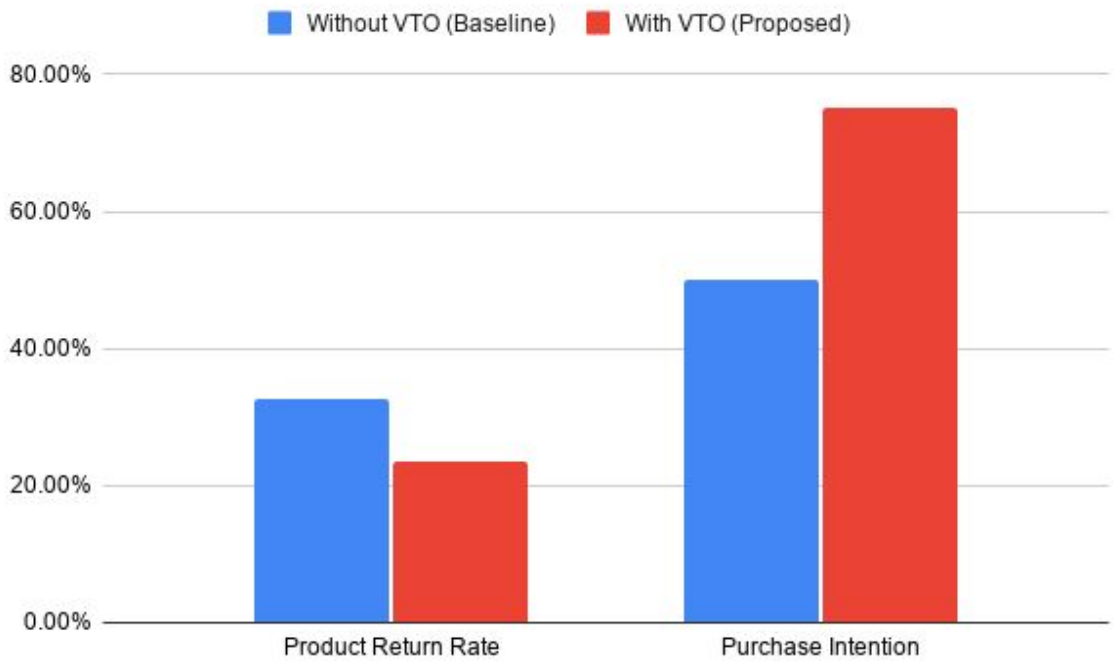
Introduction

- **The Gap:** Customers cannot try on apparel before purchasing online.
- **Industry Pain Points:** Low purchase confidence and high return rates.
- **Objective:** To create an ML-VTON system that reduces return rates and increases sales via photorealistic visualization.

Business Impact (The "Why")

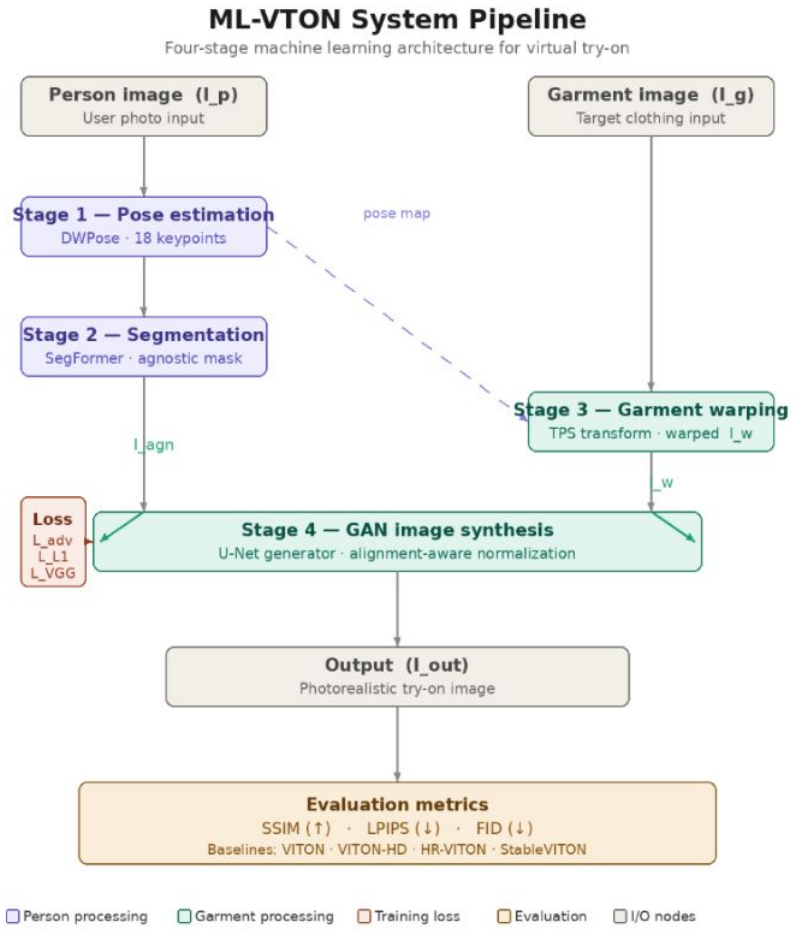
- **Key Data:** VTO implementation reduces return rates by **27%**.
- **Revenue Growth:** Average customer spending increases by **$13** per person.
- **User Intent:** Over **75% of users** are more willing to purchase with a VTO tool.



System Architecture

The 4 Stages:

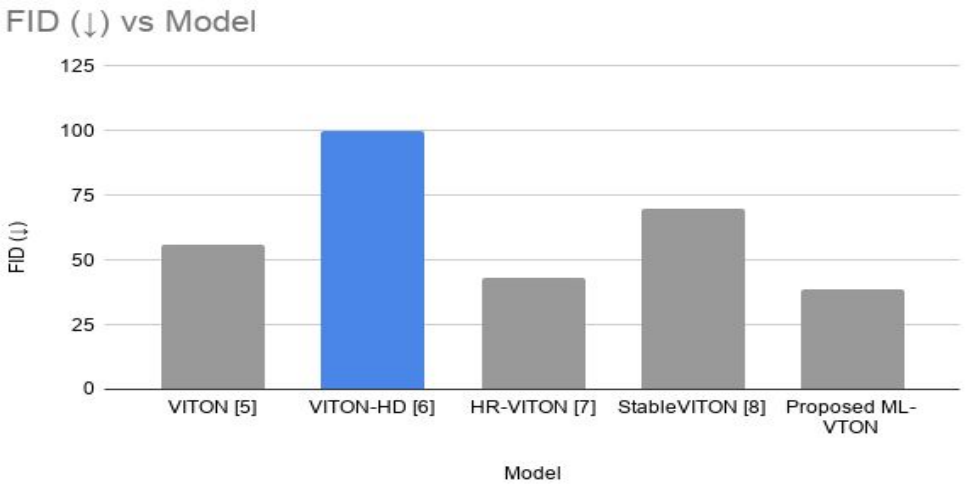
- Stage 1 - Pose Estimation: Uses DWPose to detect 18 anatomical keypoints.
- Stage 2 - Semantic Segmentation: Employs SegFormer to label pixels and create an "agnostic" mask to prevent color bleeding.
- Stage 3 - Garment Warping: Uses Thin-Plate Spline (TPS) transformation to align clothing with the user's body shape.
- Stage 4 - Image Synthesis: Utilizes a GAN (U-Net based) to generate the final photorealistic image.

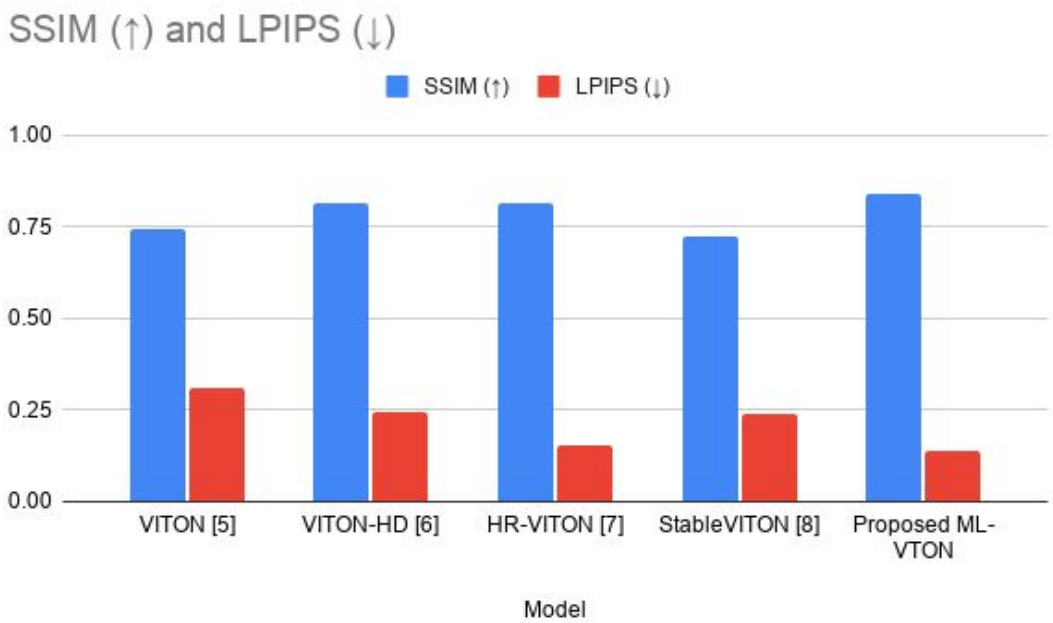


Performance Benchmarking (The Evidence)

Key Results: Our model achieves a projected SSIM of 0.841 and a FID of 38.50, outperforming baselines like StableVITON and HR-VITON.

Realism: Lower FID scores prove our system produces more realistic textures than existing high-resolution models.



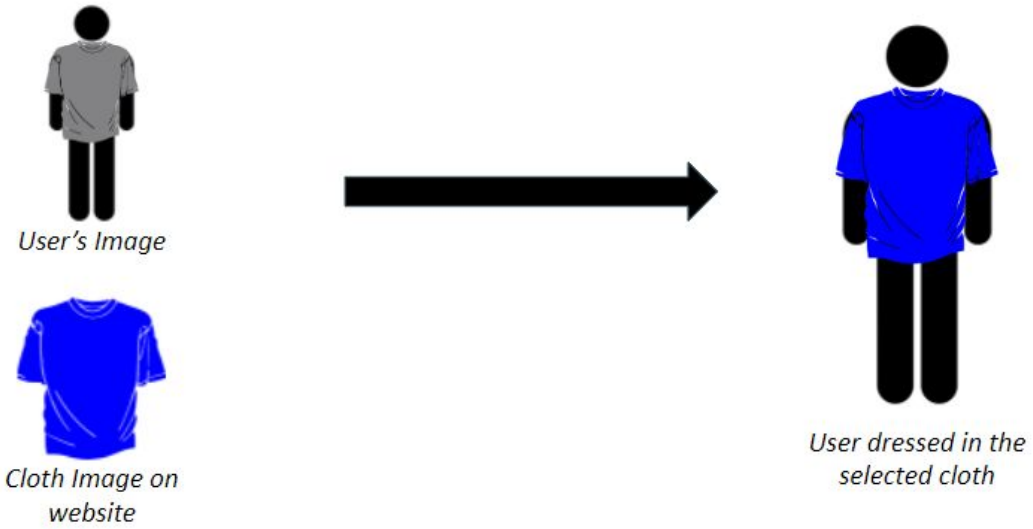


Cultural Inclusivity & Tech Stack

- Social Impact: Addresses the failure of Western-trained models on culturally complex garments like indian ethnic dresses.

- Tech Stack: Built using Python 3.10+, PyTorch (core deep learning), TensorFlow (pre-processing), and OpenCV.

- Open Access: To be released as an open RESTful API for e-commerce platforms



Conclusion & Future Work

- **Summary:** ML-VTON provides a robust, modular solution for high-fidelity virtual try-ons.
- **Future Goals:** Incorporating **fabric physics** (drape and deformation) to improve realism for loose garments.
- **Expansion:** Scaling user studies across more diverse body shapes.

Thank You